ACADEMIC WRITING INFRASTRUCTURE

ResearchGit

Version control, collaboration, and citation management for academic papers

FORMAT
Reveal.js showcase deck

FOCUS
Engineering workflows for research

PROBLEM

Academic papers still rely on fragile document workflows

Drafts split across files, folders, and email attachments

Citation edits are hard to trace and easy to break

Collaboration lacks clear history and ownership

Reviewer feedback gets buried outside the writing workflow

SOLUTION

ResearchGit adapts software version control to paper writing

Track every revision of a manuscript with commit history

Use branches for sections, experiments, or response rounds

Manage citations as structured assets instead of manual edits

Make collaboration auditable, reviewable, and reproducible

CORE FEATURES

Built for research teams, not generic source code alone

Versioned manuscripts

Restore earlier drafts, compare revisions, and preserve decision history.

Branching by section

Write Introduction, Methods, and Discussion in parallel without collisions.

Citation-aware workflow

Keep references structured and ready for BibTeX or similar formats.

Team collaboration

See who changed what, coordinate edits, and incorporate feedback cleanly.

TYPICAL WORKFLOW

Simple commands, research-specific outcomes

Initialize

```
researchgit init
```

Start a paper project with version control from day one.

Save progress

```
researchgit add
researchgit commit -m "draft methods"
```

Capture paper edits, new figures, or reference updates with intent.

Collaborate

```
researchgit push
researchgit pull
```

Sync changes through a shared repository for distributed co-authors.

Parallelize

```
researchgit branch
researchgit merge
```

Develop sections independently, then merge into the main manuscript.

BENEFITS

Why this is better than ad hoc file sharing

Safer revision history with rollback and recovery

Cleaner collaboration with fewer conflicting edits

More reliable citation and bibliography handling

Built-in backup and cloud-synced project continuity

ENGINEERING MINDSET

Make academic research flow more like disciplined software delivery

Version control + reviews

Apply pull-request style review to manuscript revisions and comments.

Agile research planning

Break work into trackable tasks, milestones, and sprint-sized goals.

Structured documentation

Keep notes, methods, and decisions in Markdown or LaTeX alongside the paper.

Automation-first reporting

Generate summaries, status updates, and reproducible outputs automatically.

AI AUTOMATION LAYER

ResearchGit can become an AI-friendly platform for academic workflows

Research acceleration

Literature search and summarization

Citation recommendations and bibliography updates

Drafting support for sections and abstracts

Consistency and plagiarism checks

Operational automation

Data collection, cleaning, and formatting

Review assistance for clarity, bias, and structure

Task prioritization and bottleneck detection

Submission and reporting pipelines

PLATFORM VIEW

What the end-to-end system could include

Submission pipeline for journals and conferences
Collaboration hub with review, task tracking, and history
AI services for analysis, writing support, and quality checks
Centralized repository for manuscripts, data, and references

ROADMAP

Build in stages, validate with real collaborators

Implement project initialization and revision tracking.

Add branching and merging flows for collaborative writing.

Integrate citation management through BibTeX-style references.

Support remote sync, review loops, and team feedback.

CLOSING

ResearchGit aims to make paper development reproducible, collaborative, and automation-ready

The goal is not just a better writing tool. It is a research workflow platform that treats papers with the same rigor modern teams apply to software.

Next step: prototype with a small research team and iterate from real usage.